

1. GENERAL

1.1 TITLE: **Artificial Intelligence And Its Application In Biomedical Engineering**

1.2 *COURSE NUMBER (if known): BM 511

1.3 CREDITS: 3L-0T-0P 9 Credits

1.4 SEMESTER-OFFERED:

1.5 PRE-REQUISITES: Basics of Electrical Engineering, Electronics Engineering and Instrumentation.

2. OBJECTIVE: The course aims to provide an understanding of the ANN, Fuzzy system and optimization algorithms.

3. COURSE TOPICS:

1. Introduction: Introduction to Artificial Neural Networks (ANN), terminology, models of neurons, topology. Activation and synaptic dynamics: Activation dynamic models, synaptic dynamic models. Types of learning, supervised and unsupervised learning, overview of learning methods: Clustering, Competitive learning, k-NN, Hebbian learning. **(6 lectures)**

2. Pattern recognition: Pattern recognition tasks: Pattern recognition problems, basic functional units. Feed forward neural networks: Analysis of pattern association networks, analysis of pattern classification networks, Feedback neural networks: Analysis of linear associative. **(4 lectures)**

3. Fuzzy System: Introduction to fuzzy system, crisp v/s fuzzy sets, fuzzy sets to fuzzy event. Fuzzy logic, Practical fuzzy measures, Fuzzy set operations, properties of fuzzy sets, fuzzification techniques, relational inference, compositional inference. Linguistic variables and logic operators, Inference using fuzzy variables, Fuzzy implication. Defuzzification. **(10 lectures)**

4. Evolutionary Computation: Introduction to Genetic Algorithm: biology, method, variants, applications. Genetic Programming: evolving computer programs, tree/linear/graph based genomes. Evolution Strategies: method, variations, optimization **(6 lectures)**

5. AI Applications: Biomedical applications of ANN: Modeling and diagnosing the cardiovascular system, Pattern recognizing of pathology images, ultrasound and magnetic resonance medical images textures analysis using ANN. **(5 lectures)**

READINGS

4.1 TEXTBOOK:

- Dmnna L. Hudson and Maurice B. Coten., “Neural Networks and Artificial Intelligence for Biomedical Engineering”, Prentice Hall of India. Pvt. Ltd., New Delhi
- D L Hudson and M E Cohen, “Neural Networks and Artificial Intelligence for Biomedical Engineering”, Prentice Hall
- Vojislav Kecman, “Learning and soft computing”, Pearson Education (Asia) Pte. Ltd.
- S. Rajasekaran, G. A. Vijayalakshmi pai, Neural networks, fuzzy logic and genetic algorithm: synthesis and applications, PHI Learning

4.2 *REFERENCE BOOKS:

S.Haykin, “Neural networks: A Comprehensive Foundation” Pearson Education (Asia) Pte. Ltd/Prentice Hall of India